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EDUCATION

Fisk University, Nashville, TN

B.S. Computer Science, B.A. Mathematics

Expected: May 2028

Presidential Scholar – GPA: 3.7/4.0

Courses: Software Design, Algorithms, Operating Systems, Computer Networks, Machine Learning, Theory of Computation, OOP

EXPERIENCE

Google

Sunnyvale, CA

Software Developer Intern

May 2025 – Aug 2025

- Built the *PRICE*Less pipeline, reducing PRICE (Production Risks in Core Engineering) assessment time from 15 days to 15 min.
- Designed a submit-time verification gate using an LLM grounded via RAG on SRE principles to flag missing or irrelevant answers.
- Automated manual emails/Buganizer follow-ups with a feedback loop that alerts owners to gaps and re-verifies on resubmit.
- Implemented the synthesis stage, merging PRICE dashboard metrics into a standardized, RAG-grounded report for SRE sign-off.
- Integrated LOAS authentication to securely enable service-to-service communications, user authorization, and internal API triggers.

Vanderbilt University

Nashville, TN

Software Developer (part-time)

Aug 2026 – Present

- Developing a Vision AI system for structured data extraction from confidential images and handwritten documents.
- Engineer a C++/OpenCV pipeline for real-time document detection, image preprocessing, and OCR/handwriting recognition.
- Integrating local VLMs with confidence validation to securely transform extracted content into structured data, entirely offline.

Software Developer Intern

May 2026 – Aug 2026

- Migrated 5 years of biosafety compliance records from a legacy system to a modern third-party platform (Onsite Systems).
- Built an ETL pipeline with dual extractors (pdfplumber, PyMuPDF) and a local LLM verifier, cutting manual data entry by 90%.
- Normalized migrated records with RapidFuzz and built a unified query interface to satisfy strict vendor API requirements.

PROJECTS

FlashFS

Mar 2026 – Present

- Built a FUSE-based distributed filesystem sharding files into content-addressed chunks across Linux nodes; cloud stores metadata only.
- Enabled direct P2P transfers across NATs using STUN/ICE and secured gRPC communication between nodes and services with mTLS.
- Added fault-tolerance mechanisms including quorum replication, WAL-backed recovery, and self-healing repair during node failures.

AutoGrocery

Jan 2026 – Mar 2026

- Architected 3 Go microservices (ordering, inventory, pricing) over gRPC, backed by PostgreSQL and Redis for distributed processing.
- Built C++ warehouse-robot workers using ZeroMQ pub-sub; exposed a REST API for authentication, order management, and billing.

AudioSync

Dec 2025 – Jan 2026

- Built a C++ multi-speaker audio sync tool for A2DP devices using concurrent playback pipelines and timestamped buffering.
- Minimized inter-speaker playback drift through clock-offset estimation and latency-compensated real-time synchronization.

TECHNICAL SKILLS

Systems: C/C++, Go, Linux, FUSE, Concurrency, Distributed Systems, Message Queues

Networking: TCP/IP, HTTP, gRPC, WebSockets, STUN/ICE, mTLS, PKI

Backend: Python, Django, SQL, ORM, PostgreSQL, MongoDB, Redis

Infrastructure & Tools: Docker, Kubernetes, AWS IoT, Git

LEADERSHIP & AWARDS

- VP, Fisk Math Club
- Finalist, Nepal Olympiad in Informatics 2022
- Semifinalist, Nepal Mathematical Olympiad 2022